Program: B.Sc. Engg. (CSE/SWE), 1st Semester | Date: October 28, 2024 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

Mid Semester Examination | Winter Semester: 2023 - 2024
Course Number: **Math 4141** | Full Marks: 160
Course Title: Geometry and Differential Calculus | Time: 2 Hours

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 4 (Four)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

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1. (a) (i). A piecewise function f be defined as follows: **[15]** (CO1) (PO1)

$$f(x)=\begin{cases}\dfrac{1}{x+2}; x<-2\\ x^2-5; -2<x\le 3\\ \sqrt{x+13}; x>3\end{cases}$$

Find $\lim\limits_{x\to -2} f(x)$, $\lim\limits_{x\to 0} f(x)$, and $\lim\limits_{x\to 3} f(x)$.

(ii). Find the values of the constants k and m, if possible, that will make the function f continuous everywhere.

$$f(x)=\begin{cases}x^2+5; x>2\\ m(x+1)+k; -1<x\le 2\\ 2x^3+x+7; x\le -1\end{cases}$$

(b) (i). Find $f'(x)$, if $f(x)=\left(\dfrac{1}{x}+\dfrac{1}{x^2}\right)\left(3x^3+27\right)$ using product rule. **[15]** (CO2) (PO1)

(ii). Find $g'(x)$, if $g(x)=\dfrac{x^3+2x^2-1}{x+5}$ using quotient rule.

(iii). Find $\dfrac{dy}{dx}$, if $y=\left(1+x^5\cot x\right)^{-8}$ using chain rule.

(c) During the first $40s$ of a rocket flight, the rocket is propelled straight up so that in t seconds it reaches a height of $s=0.3t^3$ feet. Then, **[10]** (CO2) (PO2)

(i). Find the height of the rocket travel in 40 seconds.
(ii). Find the average velocity of the rocket during the first 40 seconds.
(iii). Find the average velocity of the rocket during the first 1000 feet of its flight.
(iv). Find the instantaneous velocity of the rocket at the end of 40 seconds.

2. (a) (i). Write down the statement of Leibnitz's Theorem.

(ii). If $y = a\cos(\ln x) + b\sin(\ln x)$, then show that

$x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0.$

(iii). If $y = x^{2n}$, then show that $y_n = 2^n \{1.3.5.\cdots(2n-1)\}x^n$, where $n \in N$

(CO
(PO1

(b) Use implicit differentiation technique to find $\frac{dy}{dx}$ for a Trisectrix represented by $y^3 + yx^2 + x^2 - 3y^2 = 0$ Find an equation for the tangent line to the Trisectrix, mentioned above, at the point $(0,3)$. [7] (CO2) (PO1)

(c) A baseball diamond is a square whose sides are 27 m long. Suppose that a player running from second base to third base has a speed of 9 m/s at the instant when he is 6 m from third base. At what rate is the player's distance from home plate changing at that instant? [8] (CO2) (PO2)

3. (a) Transform the equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ to one in which there is no term involving of x, y and xy both sets of axes being rectangular. Hence, sketch the standard form of the equation. [15] (CO1) (PO1)

(b) Prove that the equation $2x^2 - 7xy + 3y^2 + x + 7y - 6 = 0$ represents a pair of straight lines. Hence, Find [25] (CO2) (PO2)

(i). the angle between the straight lines.

(ii). the equation of the bisectors of the lines.

(iii). Also, sketch the pair of straight lines and its bisectors.

4. (a) (i). Reduce the equation $f(x,y) \equiv x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$ to its standard form and find all its properties. [25] (CO2) (PO2)

(ii). Sketch the curve of the equation.

(b) Find the co-ordinates of the limiting point of the co-axial system determined by the circles $x^2 + y^2 + 4x + 2y + 5 = 0$ and $x^2 + y^2 + 2x + 4y + 7 = 0$. [15] (CO1) (PO1)

The End

$4x - 7y + 1 = 0$

$-7x + 6y + 7 = 0$

$\frac{a\cos\left(\frac{n\pi}{2} + x\right)(-1)^{n-1}(n-1)!}{x^n}$